

# Electromagnetic Waves and Colors of Light - HKDSE Physics Summary

## 1. Visible Spectrum and Colors

### 1.1 Visible Spectrum

- **White light** is a mixture of different colors
- When white light passes through a prism, it disperses into a **visible spectrum** or **color spectrum**
- Visible spectrum order (wavelength from long to short): **Red, Orange, Yellow, Green, Blue, Indigo, Violet**

### 1.2 Primary Colors

The three primary colors of visible light are:

-  **Red**
-  **Green**
-  **Blue**

**Important property:**

- All colors can be produced by mixing these three primary colors in different proportions

### 1.3 Secondary Colors

Any two primary colors mixed produce secondary colors:

| Mixture      | Secondary Color Produced                                                                           |
|--------------|----------------------------------------------------------------------------------------------------|
| Red + Green  |  <b>Yellow</b>  |
| Green + Blue |  <b>Cyan</b>    |
| Blue + Red   |  <b>Magenta</b> |

### 1.4 Colors of Objects in Colored Light

The color of an object is determined by **the colors of light it reflects**:

| Object Type         | Reflection Property | Color in White Light |
|---------------------|---------------------|----------------------|
| <b>White object</b> | Reflects all colors | White                |

|                          |                                             |                |
|--------------------------|---------------------------------------------|----------------|
| <b>Colored object</b>    | Only reflects its own color, absorbs others | Object's color |
| <b>Black/Dark object</b> | Absorbs all colors                          | Black          |

#### Example:

- Red object in white light → Reflects red, absorbs green and blue → Appears red
- Red object in green light → Absorbs green, no red to reflect → Appears black

## 2. Electromagnetic Spectrum

### 2.1 Composition of EM Spectrum

The electromagnetic spectrum consists of **visible spectrum** and **invisible spectrum**.

**Complete EM spectrum (wavelength from long to short):**

1. **Radio waves** - Longest wavelength
2. **Microwaves**
3. **Infrared**
4. **Visible light**
5. **Ultraviolet**
6. **X-rays**
7. **Gamma rays** - Shortest wavelength

**Memory Mnemonic:**

**R M I V U X G**

(Radio, Microwaves, Infrared, Visible, Ultraviolet, X-rays, Gamma)

## 3. Properties and Applications of EM Waves

### 3.1 Radio Waves

- **Wavelength range:**  
 $10^{-1}$   
 –

$$10^4$$

m (longest)

- **Source:** Radio and TV transmitters
- **Detector:** Antenna
- **Applications:**
  - Radio/TV broadcasting
  - Mobile phone communication

### 3.2 Microwaves

- **Wavelength range:**  
 $10^{-3}$   
—  
 $10^{-1}$   
m
- **Source:** Microwave transmitter
- **Detector:** Microwave receiver
- **Applications:**
  - Microwave oven
  - Satellite communication
  - Radar

### 3.3 Infrared

- **Wavelength range:**  
 $10^{-6}$   
—  
 $10^{-3}$   
m
- **Source:** Objects above 0 K (all warm objects)
- **Detector:** Skin, phototransistor
- **Applications:**
  - TV remote control
  - Camera auto-focus
  - Infrared thermometer
  - Radiant heater

- Night vision camera

### 3.4 Visible Light

- **Wavelength range:**  
 $4 \times 10^{-7}$   
–  
 $7 \times 10^{-7}$   
m
- **Source:** Hot objects, Sun
- **Detector:** Human eye
- **Applications:**
  - Lighting
  - Vision

### 3.5 Ultraviolet (UV)

- **Wavelength range:**  
 $10^{-9}$   
–  
 $10^{-8}$   
m
- **Source:** Very hot objects, UV lamps, Sun
- **Detector:** Fluorescent material
- **Applications:**
  - Sterilizing drinking water
  - Checking banknotes
  - Producing Vitamin D in skin
- **Danger:** Over-exposure may cause skin cancer

### 3.6 X-rays

- **Wavelength range:**  
 $10^{-10}$   
m
- **Source:** X-ray tube
- **Detector:** Photographic film
- **Applications:**

- See-through images (human body or luggage)
- CT scan
- Medical diagnosis

### 3.7 Gamma Rays

- **Wavelength range:**  
 $10^{-12}$   
 m (shortest)
  - **Source:** Radioactive substances
  - **Detector:** GM tube, photographic film
  - **Applications:**
    - Medical uses
    - Sterilization (killing bacteria in food)
    - Radiotherapy (killing cancer cells)
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## 4. Common Properties of EM Waves

All electromagnetic waves share the following properties:

### 4.1 Wave Nature

1.  **Transverse waves** - Vibration perpendicular to direction of propagation
2.  **Carry energy** - Can transfer energy
3.  **Travel in vacuum** - Do not require a medium
4.  **Same speed in vacuum** -  
 $c = 3 \times 10^8 \text{ m s}^{-1}$   
 (speed of light)
5.  **Slower in materials** - Different speeds in different media

### 4.2 Wave Equation

All electromagnetic waves obey the wave equation:

$$v = f\lambda$$

**Where:**

- $v$   
= wave speed (in vacuum:  
 $c = 3 \times 10^8 \text{ m s}^{-1}$   
)
- $f$   
= frequency (Hz)
- $\lambda$   
= wavelength (m)

## 4.3 Wave Phenomena

EM waves exhibit:

- **Reflection**
  - **Refraction**
  - **Diffraction**
  - **Interference**
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## 5. Relationship Between Wavelength, Frequency, and Energy

### 5.1 Inverse Relationship Between Wavelength and Frequency

In vacuum, since

$$c = f\lambda$$

:

$$f = \frac{c}{\lambda}$$

**Important rule:**

- Longer wavelength  $\rightarrow$  Lower frequency
- Shorter wavelength  $\rightarrow$  Higher frequency

### 5.2 EM Spectrum Ordering

**Wavelength (long to short):**

Radio waves > Microwaves > Infrared > Visible light > Ultraviolet > X-rays > Gamma rays

### Frequency (low to high):

Radio waves < Microwaves < Infrared < Visible light < Ultraviolet < X-rays < Gamma rays

### Energy (low to high):

Radio waves < Microwaves < Infrared < Visible light < Ultraviolet < X-rays < Gamma rays

## 5.3 Visible Light Ordering

Within visible light range (wavelength from long to short):

Red > Orange > Yellow > Green > Blue > Indigo > Violet

**Mnemonic: ROY G BIV**

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## 6. Important Concepts and Common Exam Points

### 6.1 Speed of EM Waves in Vacuum

- All EM waves have the **same speed** in vacuum
- $c = 3 \times 10^8 \text{ m s}^{-1}$
- This is the universal speed limit

### 6.2 Evidence that Light is an EM Wave

The following phenomena prove light is an electromagnetic wave:

1. ☒ Light can travel in vacuum (from Sun to Earth)
2. ☒ Light is a transverse wave
3. ☒ Light speed equals EM wave speed

### 6.3 Application Memory Techniques

**Microwave applications:**

- **Microwave oven**

- Satellite communication
- Radar

#### **Infrared applications:**

- **Remote** control
- Camera auto-focus
- Thermometer
- Night vision

#### **Ultraviolet applications:**

- **Sterilization** of drinking water
- Checking banknotes
- Producing Vitamin D

#### **X-ray applications:**

- **See-through** imaging
- CT scan

#### **Gamma ray applications:**

- **Sterilization**
  - Radiotherapy
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## **7. Safety Precautions**

### **7.1 Ultraviolet (UV) Danger**

- ⚠ Over-exposure may cause **skin cancer**
- ⚠ Should not over-use UV lamps for tanning
- ✅ Moderate exposure promotes Vitamin D production

### **7.2 X-rays and Gamma Rays**

- ⚠ Highly penetrating, damage body tissue
  - ⚠ Require professional operation
  - ✅ Medical use strictly controlled for dosage
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## 8. HKDSE Exam Key Points

### 8.1 Must Remember

1. **EM spectrum order** (wavelength long to short): Radio, Microwaves, Infrared, Visible, UV, X-rays, Gamma
2. **Primary colors**: Red, Green, Blue
3. **Secondary colors**: Yellow (Red+Green), Cyan (Green+Blue), Magenta (Blue+Red)
4. **Speed in vacuum**:  
 $c = 3 \times 10^8 \text{ m s}^{-1}$
5. **Wave equation**:  
 $v = f\lambda$

### 8.2 Common Question Types

1. **Ordering questions**: Arrange EM waves by wavelength/frequency/energy
2. **Application questions**: Identify applications of different EM waves
3. **Calculation questions**: Use  
 $v = f\lambda$   
to calculate wavelength or frequency
4. **Color mixing**: Primary and secondary color combinations
5. **Object colors**: Color of objects under different colored lights

### 8.3 Problem-Solving Tips

- **Wavelength and frequency inversely related**: Longer wavelength, lower frequency
- **Remember one wavelength range**: Visible light  $\approx 10^{-7}$  m as reference
- **Match applications to bands**: Microwave oven uses microwaves, remote uses infrared, sterilization uses UV

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## 9. Quick Memory Mnemonics

**EM Spectrum Order:**

## **R M I V U X G**

(Wavelength from long to short)

### **Visible Spectrum Order:**

## **ROY G BIV**

(Red, Orange, Yellow, Green, Blue, Indigo, Violet)

### **Primary Colors:**

**Red, Green, Blue - Primary three**

### **Secondary Colors:**

**Red plus Green makes Yellow**

**Green plus Blue makes Cyan**

**Blue plus Red makes Magenta**

### **EM Wave Properties:**

**Transverse waves carry energy through space**

**Same speed three hundred million**

**Reflect, refract, diffract, interfere**

**Wave equation  $v = f\lambda$**

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### **Study Suggestions:**

1. Memorize EM spectrum order and applications
2. Understand relationship between wavelength, frequency, and energy
3. Practice using
$$v = f\lambda$$
for calculations
4. Master primary and secondary color mixing rules
5. Understand practical applications and safety precautions of various EM waves